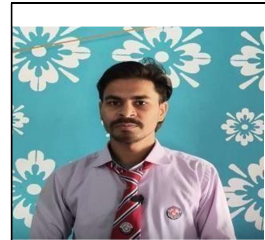


PROJECT AND TEAM INFORMATION

Project Title

Smart Warehouse Resource Management and Robot Task Scheduling System

Student / Team Information

<i>Team Name:</i>	<i>Logic Developer</i>	
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PROPOSAL DESCRIPTION

Motivation

Modern warehouses contain many customer orders, products, inventory records and daily operations. When these operations are managed manually or separately, issues like delayed order processing, incorrect inventory management, resource conflicts, and underuse of the warehouse's resources can occur.

The proposed system is designed to provide a human-centric approach to managing these activities in a single environment. Customer orders will be converted into warehouse tasks like picking, packing, and delivered. Operating System concepts will be employed to schedule these tasks, and manage context robots as shared resources.

The database will keep records of customers, products, orders, inventory, robots, and tasks. An AI module will use old order data to predict product demand and suggest task priorities. The project is useful because it demonstrates how DBMS, Operating System concepts, and AI can be integrated in a practical warehouse-management application.

State of the Art / Current solution

Modern warehouses often utilize Warehouse Management Systems and database-driven applications to store information on products, inventory, orders, storage locations and delivery. Larger facilities also tend to make use of automated machinery and robotics for transport, picking and Arranging.

While the inner workings of scheduling and resource-management are typically hidden behind the scenes, users are presented with the final product rather than the Operating System-powered processes which underpin the tasks..

The project, therefore, presents a simplified software simulation of a warehouse, which ties database management to process scheduling, resource allocation, the synchronization and deadlock handling. The system also features an AI component which analyzes historical order information for demand prediction and task-priority recommendations.

Project Goals and Milestones

This project primarily focuses on developing a Smart Warehouse Management system and Robot Task Scheduling System that connects DBMS, Operating System concepts and AI. The main objectives are:

1. To create a database for customers, products, inventory, orders, tasks, robots and execution records.
2. To convert a customer's order into warehouse tasks such as :picking, pack and delivery.
3. Implement FCFS, SJF, Priority Scheduling and Round Robin.
4. Simulate robot allocation as shared-resource management.
5. Detect deadlocks and provide a basic recovery mechanism.
6. Use historical(old) order data for demand prediction and task priority recommendations.
7. Maintain database consistency using transactions.
8. Build an administrator dashboard to monitor the system.

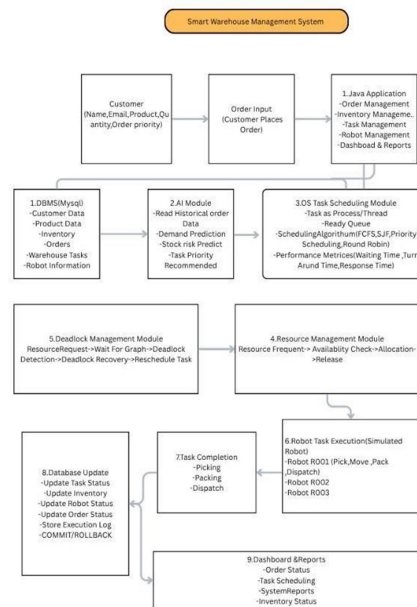
Project Approach

The proposed system will be developed as a modular software application. Each technology will be responsible for different roles and all the modules shall be integrated as one warehousemanagementsystem.

MySQL will be used to store customers, products, inventory, orders, warehouse tasks, robot resources and execution records. SQL operations such as SELECT, INSERT and UPDATE will be employed for database management. Transactions named COMMIT and ROLLBACK will be used when related database operations needed to be accomplished.

The OS simulation will represent the warehouse tasks as processes or threads. Tasks that are generated from customer orders will move onto ready queue and can be scheduled by using FCFS, SJF, Priority Scheduling and Round Robin. The Waiting time, turnaround time and response time are calculated for comparison. Robots can be represented as shared software resources. Synchronization, mutual exclusion and locks will be performed to avoid resource conflicts. A wait-for graph be implemented for deadlock detection and the affected tasks can be rescheduled. The main application will be developed by using Java. JDBC will be connect to MySQL. Python and machine-learning libraries will implement for the AI component. The AI module will provide advice, whereas, the OS module will perform the actual task scheduling. Java application serves as main application layer. It receives orders and coordinates the database. OS and AI modules. MySQL stores the warehouse information. The OS module manages the tasks such as scheduling and simulated the robot resources. The AI module analyses order historical data and provides recommendations.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The proposed project will produce a working prototype of the software platform for Smart Warehouse Resource Management and Robot Task Scheduling System. Such deliverables are going to be developed: MySQL database stocking the information on the warehouse, Java application managing orders and tasks, OS scheduling module will implement FCFS, SJF, Priority Scheduling and Round Robin algorithms., Resource allocation & synchronization module for robots, Deadlock detections and solution module, AI module for demand prediction and priority based tasks recommender, Administrator dash board manages inventory, orders, robot status and reviews scheduling outcome, projecting final documents and test outcome progress and much more.

Assumptions

The warehouse will be represented in the form of a computer software simulation rather than an actual warehouse. Robots will also be represented as software (Digital) shared resources. Product inventory and order information will be stored in MySQL.

The historical order data necessary for training the AI is prediction is either available in the database or can be generated as sample training data. The AI module will provide recommendations rather than directly control real-world robots. The topics of network communication and physical robots themselves are beyond the scope of this project.

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